



German Electricity Market  
Intelligence

Select date range

2025 / 08 / 31 – 2026 / 08 / 26



German Electricity Market Intelligence

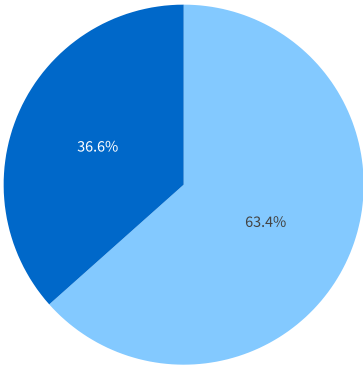
SMART hourly data • 2025-08-31 → 2026-08-26

Grid Overview

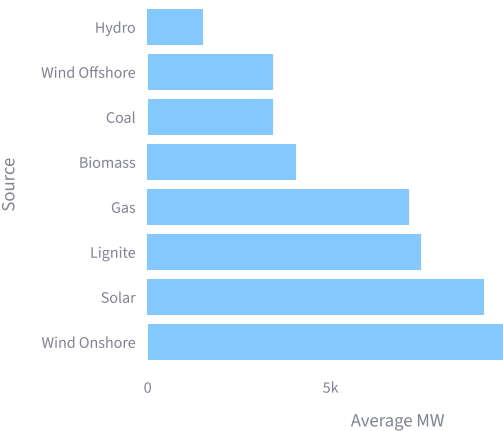
|               |                |             |                 |                      |
|---------------|----------------|-------------|-----------------|----------------------|
| Average Price | Average Demand | Peak Demand | Renewable Share | Negative Price Hours |
| €98.74...     | 53,853 ...     | 78,241 ...  | 61.1%           | 514                  |

1. Generation Mix

Renewable vs Fossil Generation



Average Generation by Source

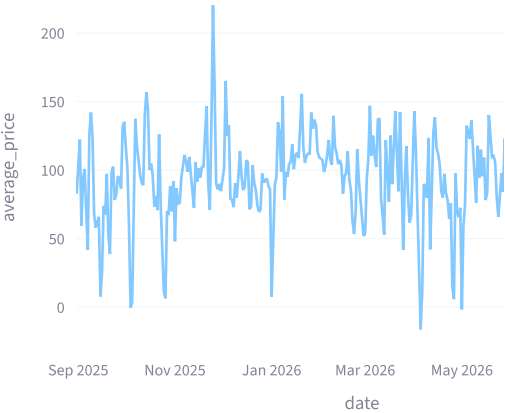


Generation Source Ranking

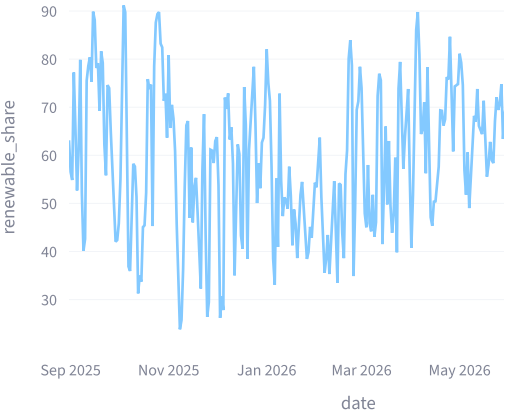
| Source        | Average MW |
|---------------|------------|
| Wind Onshore  |            |
| Solar         |            |
| Lignite       |            |
| Gas           |            |
| Biomass       |            |
| Coal          |            |
| Wind Offshore |            |
| Hydro         |            |

## 2. Market Trends

Daily Average Electricity Price

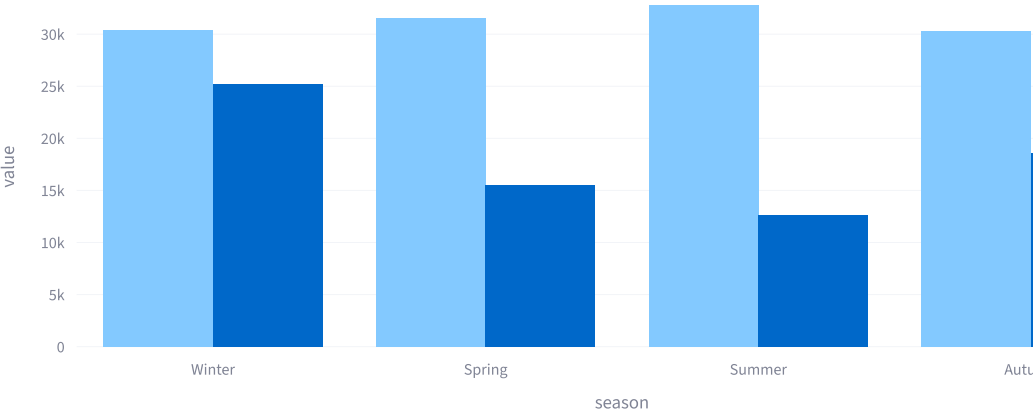


Daily Renewable Share



## 3. Seasonal Energy Analysis

Renewable vs Fossil by Season



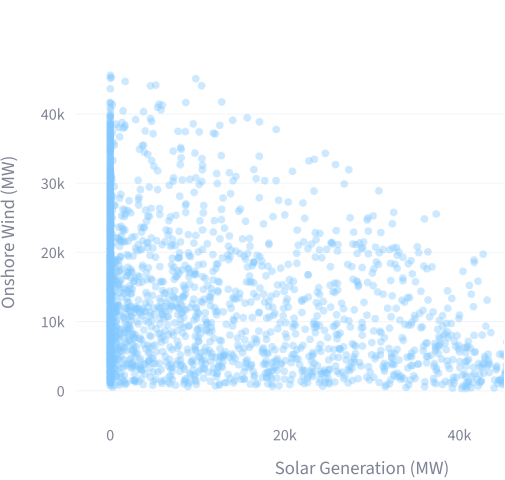
## Seasonal Performance

| season | renewable | fossil   | renewable_share | solar    | wind_onshore | wind_offshore |
|--------|-----------|----------|-----------------|----------|--------------|---------------|
| Winter | 30377.08  | 25202.03 | 53.94           | 2630.45  | 16972.93     | 5018.52       |
| Spring | 31508.37  | 15531.68 | 63.23           | 13087.46 | 10170.34     | 2574.95       |
| Summer | 32836.05  | 12626.64 | 67.38           | 15647.83 | 9388.6       | 2331.25       |
| Autumn | 30316.94  | 18533.61 | 60.14           | 5627.02  | 15275.09     | 3674.94       |

Highest renewable share: **Summer** (67.4%). Lowest average electricity price: **Autumn** (€89.87/MWh).

# 4. Solar & Wind Complementarity

Solar vs Onshore Wind



Correlation Matrix

|                  | solar_mw | wind_onshore_mw | wind_o |
|------------------|----------|-----------------|--------|
| solar_mw         | 1        | -0.295          |        |
| wind_onshore_mw  | -0.295   | 1               |        |
| wind_offshore_mw | -0.3     | 0.657           |        |

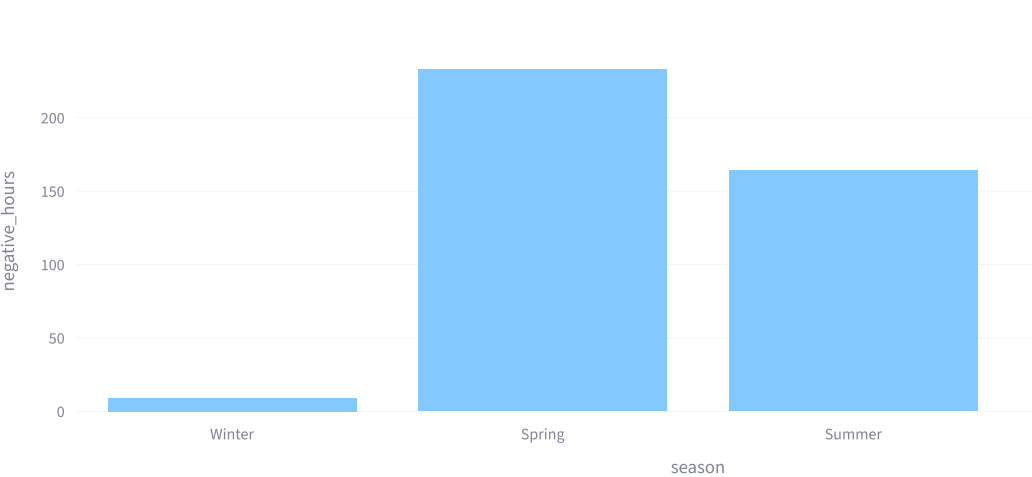
Solar ↔ Onshore Wind

-0.295

# 5. Negative Electricity Prices

| negative_price                      | average_price | renewable_generation | renewable_share |
|-------------------------------------|---------------|----------------------|-----------------|
| <input type="checkbox"/>            | 106           | 29675.6              |                 |
| <input checked="" type="checkbox"/> | -15.79        | 55930.99             | 9               |

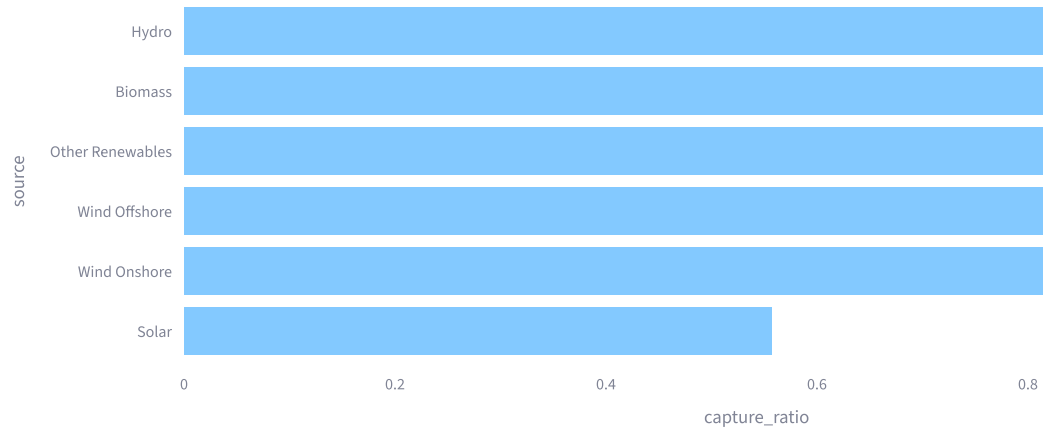
Negative-Price Hours by Season



There were 514 negative-price hours in the selected period. Average renewable generation during those hours was 55,931 MW, compared with 29,676 MW during non-negative-price hours.

# 6. Renewable Price Effectiveness

Renewable Price Capture Ratio



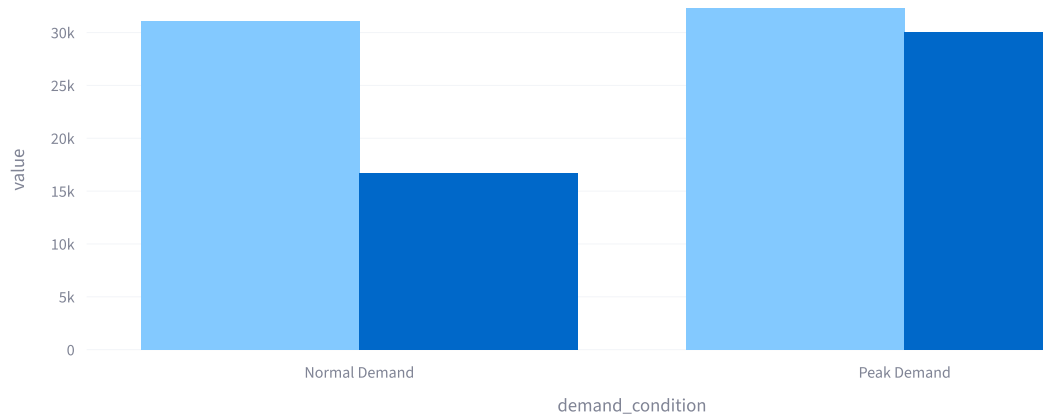
| source           | weighted_price | capture_ratio |
|------------------|----------------|---------------|
| Wind Onshore     | 84.475         |               |
| Wind Offshore    | 91.272         |               |
| Solar            | 54.977         |               |
| Biomass          | 101.55         |               |
| Hydro            | 101.842        |               |
| Other Renewables | 98.721         |               |

Highest price capture: **Hydro** (1.03). Weakest price capture: **Solar** (0.56).

## 7. Peak Demand Analysis

| demand_condition | demand   | renewable | fossil |
|------------------|----------|-----------|--------|
| Normal Demand    | 51992.59 | 31117.45  | 16684. |
| Peak Demand      | 70596.66 | 32356.42  | 30030. |

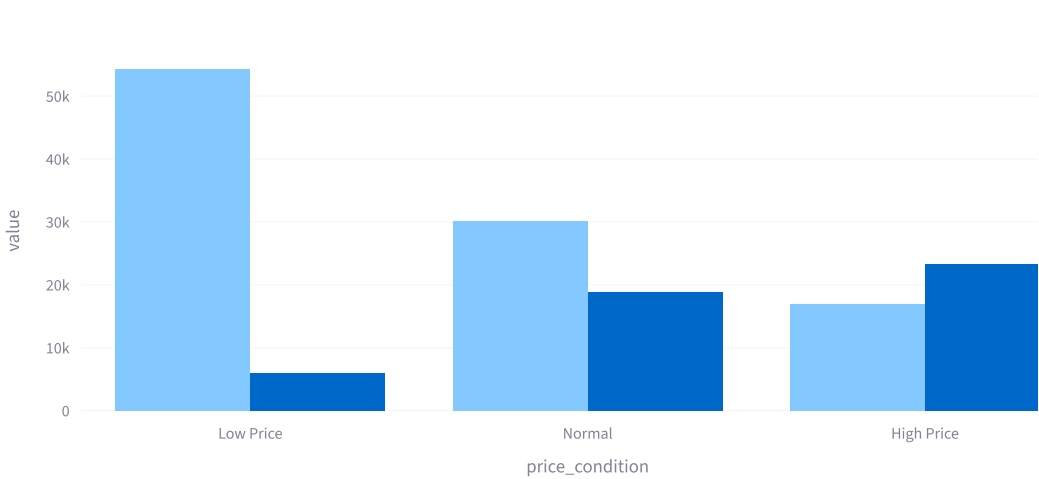
Generation During Demand Conditions



## 8. Low vs High Price Conditions

| price_condition | average_price | renewable_generation | renewable_share | fossil_generation |
|-----------------|---------------|----------------------|-----------------|-------------------|
| Low Price       | -8.18         | 54373.75             | 90.04           |                   |
| Normal          | 99.87         | 30140.15             | 59.84           | 18                |
| High Price      | 196.62        | 16917.4              | 42.38           | 25                |

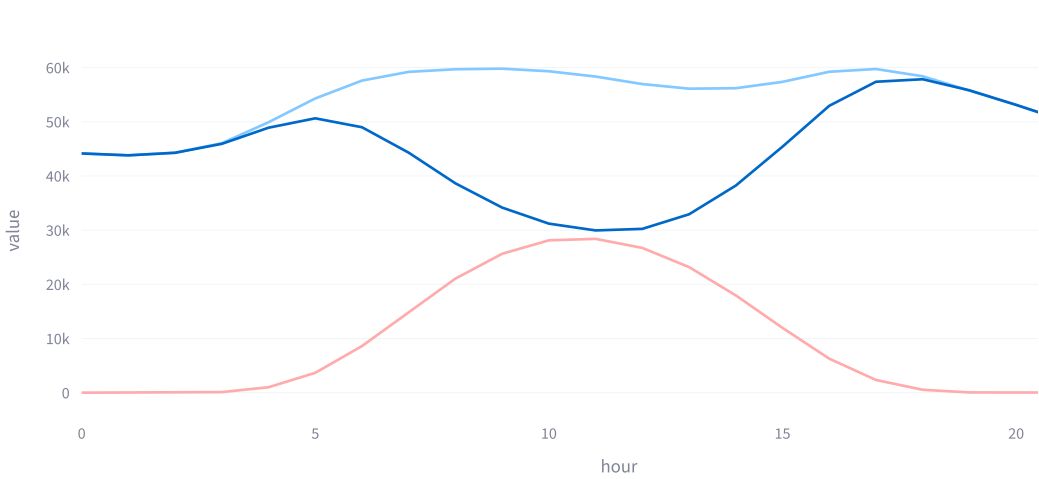
Generation During Price Conditions



## 9. Duck Curve

Average hourly demand, solar generation and demand after solar.

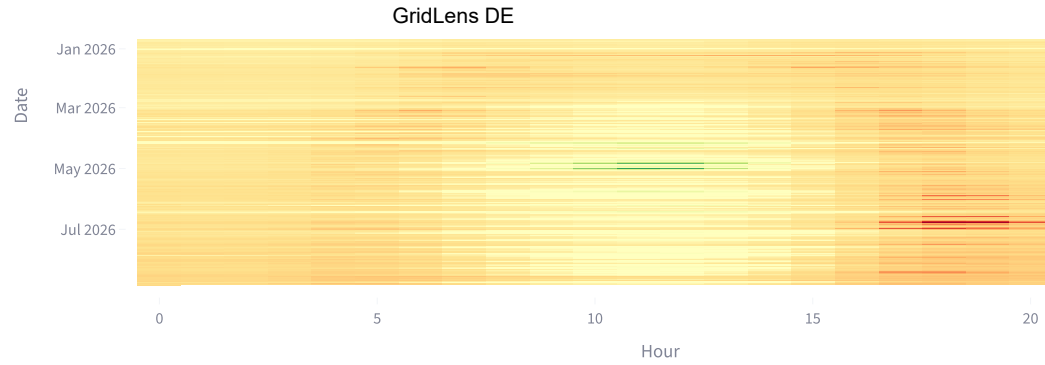
Average Daily Duck Curve



## 10. Electricity Price Heatmap

German Electricity Price Heatmap





## 11. Key Findings

- Strongest renewable season: Summer

Strongest solar season: Summer
- Strongest wind season: Winter

Lowest-price season: Autumn

## Ask GridLens

Ask questions about the German electricity-market data.

 what happens in summer

 **Summer snapshot**

- **Renewable generation:** 32,836 MW (highest of the year).
- **Fossil generation:** 12,627 MW (lowest of the year).
- **Renewable share:** 67.4 % of total generation.
- **Solar output:** 15,648 MW (highest solar output).
- **Wind on-shore:** 9,389 MW; **wind offshore:** 2,331 MW.
- **Average price:** €113.87/MWh (above the yearly average).
- **Demand:** 49,969 MW (lower than the yearly average).
- **Negative-price hours:** 164 (more than in winter, but fewer than in spring).

**What it means**

In summer the market is dominated by renewables, especially solar, which drives up the renewable share to two-thirds of total generation. Because demand is lower and renewable output high, prices rise above the yearly average, but negative-price events still occur (164 hours). The strong solar output also weakens the correlation between renewable generation and price (overall correlation is  $-0.701$ ).

## 12. Data Explorer

| timestamp           | consumption_mw | renewable_mw | fossil_mw | solar_mw | wind_onshore_mw | wind_offst |
|---------------------|----------------|--------------|-----------|----------|-----------------|------------|
| 2026-08-21 21:00:00 | 40808.5        | 13523.9      | 17364.82  | 0        | 5470.62         |            |
| 2026-08-21 22:00:00 | 38612.64       | 13913.88     | 16800.6   | 0        | 5802.63         |            |
| 2026-08-21 23:00:00 | 37406.86       | 15762.83     | 15974.07  | 0        | 6705.43         |            |
| 2026-08-22 00:00:00 | 37577          | 16947.87     | 14759.84  | 0        | 7191.22         |            |
| 2026-08-22 01:00:00 | 36975.04       | 17401.37     | 14531.41  | 0        | 7479.28         |            |
| 2026-08-22 02:00:00 | 37431.52       | 18345.43     | 14293.98  | 0        | 8221.57         |            |
| 2026-08-22 03:00:00 | 39210.23       | 20103.67     | 13946.68  | 0.04     | 9147.24         |            |
| 2026-08-22 04:00:00 | 39895.36       | 22388.96     | 12874.4   | 432.56   | 10049.36        |            |
| 2026-08-22 05:00:00 | 42110.11       | 26836.48     | 12320.2   | 4142.31  | 10232.61        |            |
| 2026-08-22 06:00:00 | 44491.87       | 33629.11     | 10398.42  | 11469.24 | 9827.74         |            |
|                     |                |              |           |          |                 |            |

GridLens DE | SMARD data | Python + Pandas + Plotly + Streamlit

Ask about the German electricity market...

↑